apnea, a respiratory condition characterised by snoring, breathing difficulties and poor-quality sleep. Then there are a few rare conditions like narcolepsy (which makes people drowsy in the day) and hypersomnia (excessive sleepiness).

Lack of sleep, or 'sleep debt or deficit' as it is termed, where an individual slips up on fulfilling the body's requirement for rest, can have an adverse effect on every organ of the body. Poor sleep is reported to cause issues related to the skin, heart, digestion, fertility, cognition and emotions as well as diabetes and can even affect the way you walk. In everyday life, poor sleep can affect your academic performance, workplace productivity and even lead to road accidents if you fall asleep at the wheel.



**"Just like poor mental health can affect sleep, poor sleep can lead to mental health ills like anxiety, depression and cognitive impairment"**

**DR SATISH RAMAIAH**
***Executive Director, Sukoon Health***

## What happens when we sleep?

Our sleep usually follows a circadian rhythm, which responds to light cues. As night falls and your body senses that light is reducing, it starts producing more of a hormone called melatonin which helps you fall asleep. Research also points towards humans having a 'sleep drive', much like the desire for water or food, which on reaching a certain level, automatically makes us feel sleepy.

"Sleep is not a passive process as we previously thought," says Dr Padma Sundaram, a pulmonologist and sleep specialist at Fortis Hospital, Cunningham Road, Bengaluru. "While your body may appear immobile, internally, a lot is going on. Sleep is also made up of various stages and we need to go through all of them in order to wake up rested and refreshed." When you fall asleep, you enter a sleep cycle with various sequential stages. It begins with non-rapid eye movement (NREM) sleep, which itself has three phases. Stage one of NREM, the lightest form of sleep, usually lasts only a few minutes. Stage two of NREM is still classified as light sleep but studies show that brain activity starts to slow down, save for short bursts of neural activity, believed to be your brain organising memories and thoughts from the day just gone by. This stage takes up the most space in our sleep and one goes through multiple rounds of it. The deepest stage is stage 3 of NREM, which is such deep sleep that it might be difficult to wake someone from it—and they would probably awaken confused. In this deep sleep, your body works to repair injuries, build your immunity and regulate cognitive abilities and memory. "You need deep sleep in sufficient amounts in order to wake up feeling rested," adds Dr Sundaram. The body returns to stage 2 NREM sleep before it enters rapid eye movement or REM sleep. This is the stage where humans dream and their brain scans reveal activity that looks similar to what happens in wakefulness. Stage 3 NREM and REM sleep usually take up 25 per cent each of sleep time. This cycle between the three stages of NREM and REM happens multiple times over night, each cycle taking 90 to 120 minutes.

## What is disturbing our sleep?

As it turns out, early to bed and early to rise is not so important as being consistent about sleep timings and completing the necessary quota of sleep. "You can be a night owl or a morning lark, but you need to maintain your body's natural rhythm. Constantly shifting your sleep timings messes up your clock," says Himani Dalmia, infant and child sleep specialist certified by the Institute of Sensitive Sleep in Australia and co-author of the book, *Sleeping Like a Baby*. Equally important is not scrimping on the number of hours we sleep. Unfortunately, as Dr Suri explains, this can be a struggle. "We continue to get up early in the morning to reach school or office. However, we sleep late due to a combination of long work hours, social media and 24/7 home entertainment on OTT platforms. The net result is a decrease in the quantity of sleep. Many people also sleep less on weekdays and try to compensate for it by sleeping extra on weekends. However, research has shown that such irregular sleep patterns do not leave you feeling refreshed."

Exactly what Mumbai-based Rouhan Sharma, 47, complains of. "I get about 5-7 hours of sleep and I find it difficult to maintain regular timings. Most days I don't feel rested when I wake up. I could have spent more quality time with the wife and kids during the weekends than I now do, instead of trying to catch up on lost sleep," he says. This is something that also cropped up in a 2024 study by homegrown D2C sleep and home solutions provider Wakefit of the sleeping habits of over 10,000 people across Indian cities—almost half the people were waking up feeling tired.

The same study also points to another culprit that is rob-